Weekly Progress Report: Brain Tumor Detection System

Advancing deep learning architectures for reliable tumor classification in neuroimaging



Week's Strategic Objectives

01

Model Comparison

Systematic evaluation of previous vs. current CNN architectures using standardized metrics

02

Performance Analysis

Comprehensive assessment of accuracy, generalization capability, and overfitting indicators

03

Architecture Refinement

Enhanced CNN design for superior feature extraction and pattern recognition

04

Evaluation Pipeline

Improved testing protocols for binary classification and multi-class tumor categorization

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Key Accomplishments

Model Development

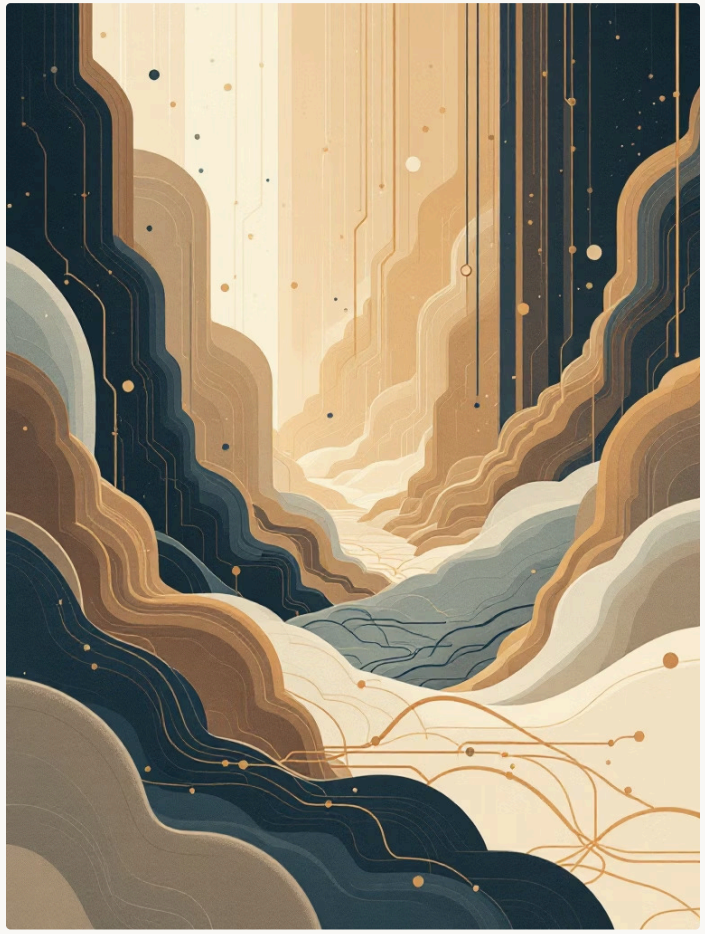
- Trained and validated baseline CNN architecture
- Developed improved model with enhanced regularization
- Established comprehensive evaluation framework

Performance Enhancement

- Reduced overfitting through architectural modifications
- Achieved stable validation accuracy
- Improved generalization to unseen data

Visualization Integration

- Progress on Grad-CAM implementation
- Frontend integration for interpretability



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Previous Model Architecture



Basic CNN Design

Simple convolutional architecture without advanced regularization techniques



High Training Accuracy

Achieved ~97% training accuracy through pattern memorization



Overfitting Issues

Significant gap between training and validation performance indicated poor generalization



Unstable Validation

Inconsistent validation accuracy across training epochs

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Current Model Architecture

Input Layer

MRI Image (224×224×3) - Standardized dimensions for consistent processing

Convolutional Block 1

32 filters, 3×3 kernel, ReLU activation, Max Pooling (2×2)

Convolutional Block 2

64 filters, 3×3 kernel, ReLU activation, Max Pooling (2×2)

Flatten Layer

Converts multi-dimensional feature maps to 1D vector

Dense Layer

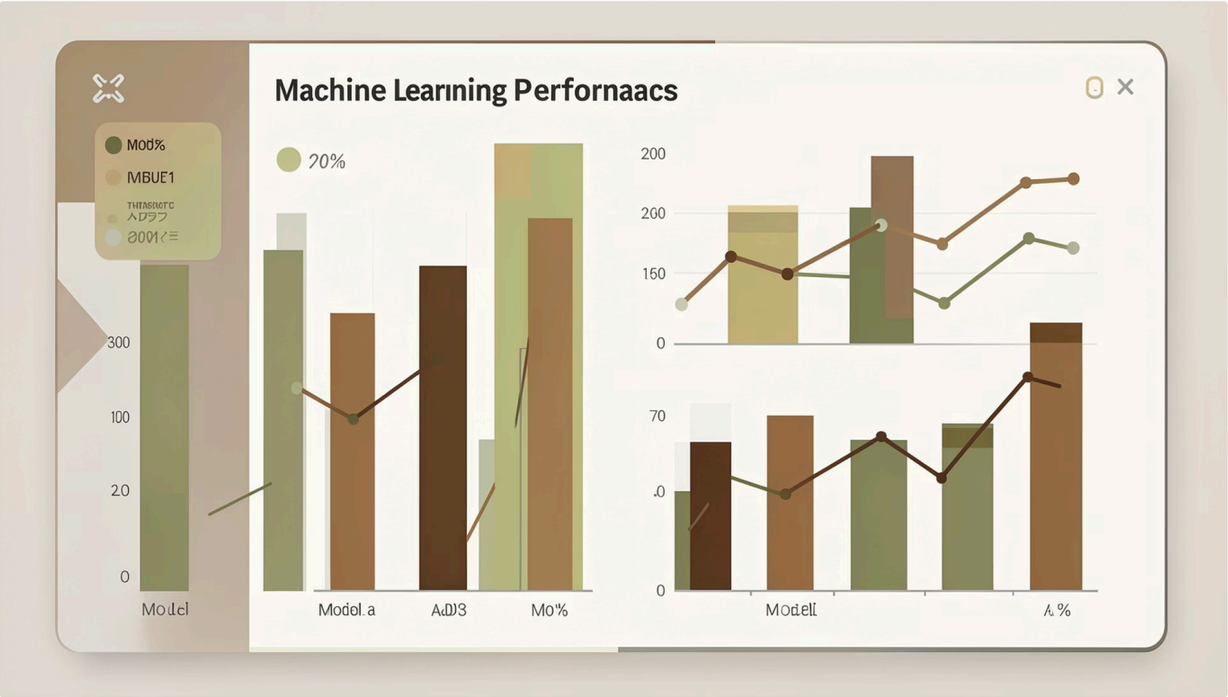
128 neurons with ReLU activation for high-level feature learning

Output Layer

Softmax activation for probabilistic classification

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Performance Comparison Analysis



Key Findings

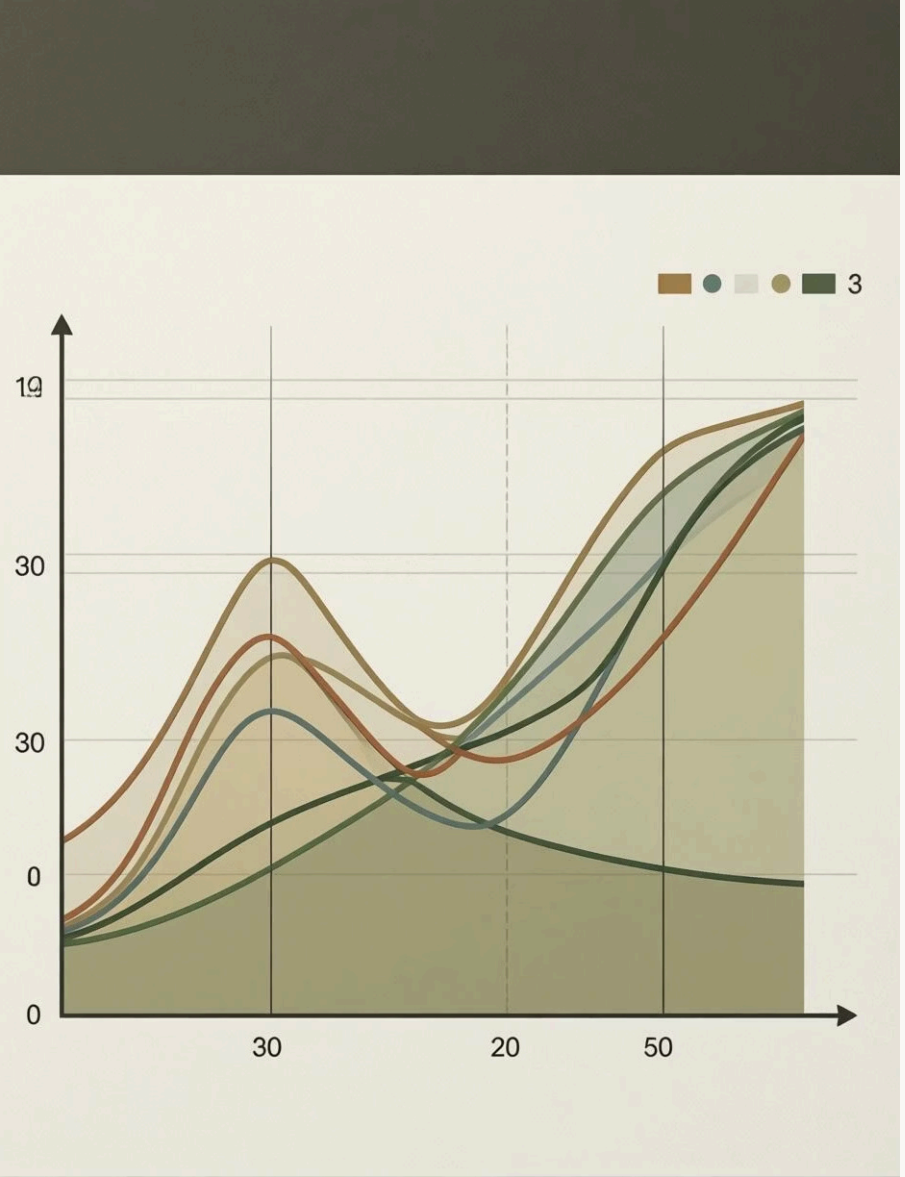
The improved model demonstrates superior real-world applicability despite slightly lower training accuracy.

Validation Stability

Current model shows consistent validation performance, indicating reliable generalization to unseen data.

Metric	Previous Model	Current Model
Training Accuracy	~97%	~96%
Validation Accuracy	Unstable	Stable
Overfitting	High	Reduced
Generalization	Weak	Improved
Reliability	Moderate	High

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Visual Evaluation: Training vs. Validation

Previous Model

Large gap between training and validation curves

✓ Significant overfitting

Current Model

Smaller gap between training and validation curves

✓ Better generalization

Key Insight: The ~96% accuracy model achieves better real-world performance through enhanced generalization, making it more suitable for clinical deployment.

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Techniques Applied & Challenges



Architecture Design

Structured convolutional blocks with systematic pooling



Enhanced Evaluation

Epoch-level monitoring of training vs. validation metrics



Model Interpretability

Grad-CAM integration for visual explanations

Challenges Encountered

Technical Issues

- Overfitting in initial model
- Missing model files (.pth)

Integration Challenges

- Frontend debugging
- Multi-class evaluation

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Key Insights & Implications

Accuracy vs. Reliability Trade-off

Slightly lower training accuracy (~96% vs. ~97%) resulted in superior validation performance and real-world applicability

Generalization Matters

The improved model learned meaningful patterns rather than memorizing training data, evidenced by reduced training-validation gap

Clinical Readiness

Stable validation accuracy and enhanced interpretability (via Grad-CAM) make current model suitable for deployment in clinical settings

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Next Steps & Recommendations

Complete Grad-CAM Integration

Finalize frontend visualization for clinical interpretability

1

Cross-Validation Testing

Validate model performance on external datasets

2

Clinical Deployment

Prepare for integration into radiology workflow systems

3

Model Strengths

- Robust generalization
- Stable validation performance
- Reduced overfitting
- Enhanced interpretability

Deployment Status

The improved CNN architecture demonstrates sufficient reliability for clinical evaluation and represents a significant advancement over the baseline model.

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